

BANARAS HINDU UNIVERSITYB.Sc. (Hons.) Semester III Examination, 2016-17

Paper No. STB-301: Probability Theory and Statistical Methods

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Explain the concept of sampling distribution with a suitable example. Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n from a population with finite mean μ and variance σ^2 . Show that:

$$E(\bar{X}) = \mu \quad \text{and} \quad V(\bar{X}) = \frac{\sigma^2}{n}$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean.

- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a Bernoulli distribution with parameter p . Show that the sampling distribution of $S = \sum_{i=1}^n X_i$ is a binomial distribution.
-

Question 2.

- (a) Define the probabilities of Type I and Type II errors. A sample of size 1 is taken from an exponential distribution with parameter λ . To test $H_0 : \lambda = 1$ against $H_1 : \lambda > 1$, suppose the critical region is considered as $x > 2$. Find the probabilities of Type I and Type II errors.
- (b) Define the null and alternative hypotheses with examples. Explain with illustrations how to specify the null and alternative hypotheses in a hypothesis testing problem.
-

Question 3.

- (a) Write the density function of the t -distribution. Mention its distributional properties. Also, define Student's t -statistic and its use.
- (b) What is F -distribution? Explain the testing procedure for testing the equality of two variances of two independent normal populations.
-

Question 4.

- (a) State the weak law of large numbers (WLLN). Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with a common Bernoulli distribution with parameter p . Show that the sequence of sample means $\{\bar{X}_n\}$ obeys the WLLN.
-

- (b) State the central limit theorem (CLT). Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n from a population with finite mean 5 and variance 16. When n is large, say $n = 40$, obtain the probability $P[\bar{X} < 5]$.
-

Question 5.

- (a) Define order statistics with examples in real-life applications. Show that the r -th order statistic from a uniform distribution $U[0, 1]$ follows a Beta distribution.
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and variance σ^2 . Show that the sampling distribution of the sample mean $\bar{X}$ is also a normal distribution.
-

Question 6.

- (a) What are the advantages and disadvantages of non-parametric tests over parametric tests?
- (b) Write the Mann-Whitney-Wilcoxon test statistic. To compare the mileage distributions of two brands of tires, the following mileages (in 1000 miles) were obtained for eight tires of each brand:

Brand A	32.1	20.6	17.8	28.4	19.6	21.4	19.9	30.1
Brand B	19.8	27.6	30.8	27.6	34.1	18.7	16.9	17.9

Test the null hypothesis that the two samples come from the same population using the Mann-Whitney-Wilcoxon test.

Question 7. Write short notes on any three of the following:

- (a) Uses of χ^2 (chi-square) distribution.
- (b) Median test.
- (c) p -value.
- (d) Test of independence of two attributes.
- (e) Fisher's Z transformation.
-

Question 8.

- (a) Explain the procedure of testing for the significance of the sample correlation coefficient r in random sampling from a bivariate normal distribution.
-

- (b) Nine students held their breath, once after breathing normally and relaxing for one minute, and once after hyperventilating for one minute. The table indicates how long (in seconds) they were able to hold their breath:

Subject	A	B	C	D	E	F	G	H	I
Normal	56.5	65.0	65.0	50.0	25.0	78.0	71.0	44.0	35.0
Hyperventilated	87.0	91.0	85.0	91.0	75.0	28.0	122.0	66.0	38.0

Mentioning the null and alternative hypotheses clearly, test the significance of the correlation coefficient between the two breath-holding times. (Given $t_{0.05,7} = 2.365$ and $t_{0.05,8} = 2.306$).